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Soluble expression of protein using peptide tag

Abstract

By linking a peptide tag having the amino acid sequence X(PY).sub.qPZ to a protein of interest, and then expressing the protein, the protein of interest is efficiently accumulated in a soluble fraction, where P is proline; X is an amino acid sequence composed of 0 to 5 amino acids independently selected from R, G, S, K, T, L, N, Q, and M; Y is an amino acid sequence composed of 1 to 4 amino acids independently selected from R, G, K, T, L, N, Q, and M; q is an integer 1 to 10; and Z is an amino acid sequence composed of 0 to 10 amino acids independently selected from R, G, S, K, T, L, N, Q, and M; where the amino acid sequence of the peptide tag contains three or more Qs, Ms, Ls, Ns, and/or Ts in total.

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Background/Summary

REFERENCE TO A SEQUENCE LISTING

(1) In accordance with 37 CFR § 1.831-1835 and 37 CFR § 1.77 (b) (5), the specification makes reference to a Sequence Listing submitted electronically as a .txt file named "535637US_012822.txt". This .txt file was generated on Jan. 28, 2022 and is 56,004 bytes in size. The Sequence Listing is already of record and has a USPTO Filing/Load Date according to Patent Center of Mar. 14, 2022. The entire contents of the Sequence Listing are hereby incorporated by reference.

TECHNICAL FIELD

(2) The present invention relates to a technology for producing a recombinant protein, more specifically, a technology for allowing efficient expression of a recombinant protein in a soluble fraction.

BACKGROUND ART

(3) Advancement of the gene recombination technique has enabled heterologous expression of useful proteins derived from higher organisms, utilizing *E. coli*, yeast, or the like. Such advancement, today, has rendered production of useful proteins by heterologous expression available as a common technique. For improvement of expression of useful proteins and their amounts accumulated, studies have been carried out in relation to selection of promoters and terminators, utilization of translational enhancers, codon modification of transgenes, intracellular transport and localization of proteins, and the like. In some other techniques that have been developed, expression of a useful protein is improved by linking a peptide tag thereto. For example, Patent Document 1 discloses that, by linking an ENTROPIC BRISTLE DOMAIN (EBD) peptide to a useful protein, and then expressing the resulting protein, improved expression of the useful protein in a soluble fraction can be achieved. However, since the EBD peptide contains serine between prolines, total protein production may be low.

PRIOR ART DOCUMENT

Patent Document

(4) [Patent Document 1] US20090137004 A

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

(5) An object of the present invention is to allow efficient production of a protein of interest in a soluble fraction, in the recombination protein production technique.

Means for Solving the Problems

(6) In order to solve the above problem, the present inventors intensively studied. As a result, the present inventors found that use of a peptide tag having the amino acid sequence shown below enables remarkable improvement of the expression level of a protein of interest in a soluble fraction. More specifically, by intensively modifying the amino acid composition and the sequence of the peptide tag to be linked, the expression level of the protein of interest in the soluble fraction was successfully and dramatically improved while the expression level of the protein of interest was maintained. The present invention was completed based on such findings.

(7) The present invention can be summarized as follows.

(8) [1] A soluble fraction prepared by an expression system comprising a polynucleotide introduced therein, the polynucleotide encoding a fusion protein containing:

(9) a protein of interest; and a peptide tag linked to the protein of interest and containing the following amino acid sequence:

X(PY).sub.qPZ wherein P represents proline; X represents an amino acid sequence composed of 0 to 5 amino acids independently selected from the group consisting of arginine (R), glycine (G),

serine (S), lysine (K), threonine (T), leucine (L), asparagine (N), glutamine (Q), and methionine (M); Y represents an amino acid sequence composed of 1 to 4 amino acids independently selected from the group consisting of R, G, K, T, L, N, Q, and M; q represents an integer of 1 to 10; and Z represents an amino acid sequence composed of 0 to 10 amino acids independently selected from the group consisting of R, G, S, K, T, L, N, Q, and M; with the proviso that the amino acid sequence of the peptide tag contains three or more Q('s), M('s), L('s), N('s), and/or T('s) in total; the soluble fraction comprising the fusion protein produced and accumulated from the polynucleotide.
[2] The soluble fraction according to [1], wherein PY is one or more selected from PGQ, PGM, PGT, PGL, PQQ, PGN, PGQG, PGMG, PGTG, PGLG, PGNG, and PQQQ.
[3] The soluble fraction according to [1] or [2], wherein the peptide tag has a length of 10 to 30 amino acids.
[4] The soluble fraction according to any one of [1] to [3], wherein the peptide tag has the amino acid sequence of SEQ ID NO:7, 10, 12, 15, 17, 19, 21, or 23.
[5] The soluble fraction according to any one of [1] to [4], wherein the protein of interest is an enzyme.
[6] The soluble fraction according to any one of [1] to [5], wherein the fusion protein contains a secretion signal.
[7] The soluble fraction according to any one of [1] to [6], wherein the peptide tag is linked to the C-terminal side of the protein of interest.
[8] A method of producing a fusion protein, the method comprising collecting the soluble fraction according to any one of [1] to [7], and extracting the fusion protein.
[9] An expression system comprising a polynucleotide introduced therein, the polynucleotide encoding a fusion protein containing: a protein of interest; and a peptide tag linked to the protein of interest and containing the following amino acid sequence:
X(PY).sub.qPZ wherein P represents proline; X represents an amino acid sequence composed of 0 to 5 amino acids independently selected from the group consisting of arginine (R), glycine (G), serine (S), lysine (K), threonine (T), leucine (L), asparagine (N), glutamine (Q), and methionine (M); Y represents an amino acid sequence composed of 1 to 4 amino acids independently selected from the group consisting of R, G, K, T, L, N, Q, and M; q represents an integer of 1 to 10; and Z represents an amino acid sequence composed of 0 to 10 amino acids independently selected from the group consisting of R, G, S, K, T, L, N, Q, and M; with the proviso that the amino acid sequence of the peptide tag contains three or more Q('s), M('s), L('s), N('s), and/or T('s) in total; wherein said expression system produces the fusion protein without denaturation.
[10] A solution containing the nondenatured fusion protein produced from the expression system according to [9].
[11] A method of increasing efficiency of material production by a metabolic system involving enzyme, the method comprising: producing the fusion protein, whose protein of interest is an enzyme, as a nondenatured protein using the expression system according to [9]; and carrying out substrate conversion reaction using the resulting nondenatured enzyme fusion protein.
[12] A fusion protein comprising: a protein of interest; and a peptide tag linked to the protein of interest and containing the following amino acid sequence:
X(PY).sub.qPZ wherein P represents proline; X represents an amino acid sequence composed of 0 to 5 amino acids independently selected from the group consisting of arginine (R), glycine (G), serine (S), lysine (K), threonine (T), leucine (L), asparagine (N), glutamine (Q), and methionine (M); Y represents an amino acid sequence composed of 1 to 4 amino acids independently selected from the group consisting of R, G, K, T, L, N, Q, and M; q represents an integer of 1 to 10; and Z represents an amino acid sequence composed of 0 to 10 amino acids independently selected from the group consisting of R, G, S, K, T, L, N, Q, and M; with the proviso that the amino acid sequence of the peptide tag contains three or more Q('s), M('s), L('s), N('s), and/or T('s) in total, and

contains at least one M.
[13] The fusion protein according to [12], wherein PY is one or more selected from PGM and PGMG.
[14] The fusion protein according to or [13], wherein the peptide tag has a length of 10 to 30 amino acids.
[15] The fusion protein according to any one of to [14], wherein the peptide tag has the amino acid sequence of SEQ ID NO:7, 10, 12, 15, 17, 19, 21, or 23.
[16] The fusion protein according to any one of to [15], wherein the protein of interest is an enzyme.
[17] The fusion protein according to any one of to [15], wherein the fusion protein contains a secretion signal.
[18] The fusion protein according to any one of to [17], wherein the peptide tag is linked to the C-terminal side of the protein of interest.
[19] A polynucleotide encoding the fusion protein according to any one of to [18].
[20] A recombinant vector comprising the polynucleotide according to [19].
Effect of the Invention
(10) According to the present invention, linking of a peptide tag containing a particular sequence enables improvement of the expression efficiency of a protein of interest and improvement of the amount of the protein accumulated in a soluble fraction, so that separation and purification of the protein of interest can be easily carried out. In cases where the protein of interest contains a secretion signal sequence, the efficiency of secretory production of the protein of interest into the medium can also be improved.
(11) In cases where the protein of interest is an enzyme, a solution containing a nondenatured enzyme fusion protein can be easily prepared from a medium, cell homogenate, or the like, and can be applied to enzymatic reaction by a metabolic system involving the enzyme, so that the enzyme can contribute to material production utilizing an efficient conversion reaction of a substrate.
(12) Moreover, since the enzyme introduced is less likely to aggregate or to become insoluble in the cell, the enzyme can efficiently contribute also to intracellular enzymatic reaction.
(13) Unlike the peptide tag described in Patent Document 1, the peptide tag to be used in the present invention does not contain serine between prolines. Thus, the peptide tag to be used in the present invention can be expected to improve production of the protein in a soluble fraction. The improvement of expression of the fusion protein in the soluble fraction can be achieved also in cases where the peptide tag is linked to the C-terminal side of the protein of interest.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a diagram illustrating a construction procedure for a plasmid for expression, in *Brevibacillus*, of fusion proteins having various peptide tags linked to the N-terminus of esterase.
(2) FIG. **2** is a diagram illustrating a construction procedure for a plasmid for expression, in *Brevibacillus*, of fusion proteins having various peptide tags linked to the C-terminus of esterase.
(3) FIG. **3** is a graph illustrating comparison of the expression level of esterase having each peptide tag linked to the N-terminal side thereof, in a soluble fraction.
(4) FIG. **4** is a graph illustrating comparison of the expression level of esterase having each peptide tag linked to the C-terminal side thereof, in a soluble fraction.
(5) FIG. **5** is a graph illustrating the result of comparison of the activity of esterase having a tag (A: PX12-32, B: PX12-34) added thereto. S represents a fraction secreted into the medium, and P represents a soluble fraction in the cell.
(6) FIG. **6** is a graph illustrating comparison of the expression level of human growth hormone

having various peptide tags linked to the N-terminal side thereof, which comparison was made for each fraction (yeast).

EMBODIMENTS FOR CARRYING OUT THE INVENTION

(7) In an expression system comprising a polynucleotide introduced therein encoding a fusion protein containing: a protein of interest; and a peptide tag linked to the protein of interest and containing the following amino acid sequence; the fusion protein can be produced from the polynucleotide and accumulated in a soluble fraction. After collecting the soluble fraction, the fusion protein can be extracted.

(8) The peptide tag used in the present invention has the following sequence.

X(PY).sub.qPZ

(9) X represents an amino acid sequence composed of 0 to 5 amino acids independently selected from the group consisting of arginine (R), glycine (G), serine (S), lysine (K), threonine (T), leucine (L), asparagine (N), glutamine (Q), and methionine (M).

(10) When X is 0 amino acid, the amino acid at the N-terminus of the peptide tag is P.

(11) When X is 1 amino acid, the amino acid at the N-terminus of the peptide tag is selected from R, G, S, K, T, L, N, Q, and M.

(12) When X is 1 amino acid, it is preferably Q, L, N, M, or T, more preferably Q, M, or T.

(13) When X is 2 to 5 amino acids, the 2 to 5 X's may be either the same amino acid residues or different amino acid residues selected from R, G, S, K, T, L, N, Q, and M.

(14) The number of X's is preferably 1 to 5, more preferably 2 to 5, still more preferably 2 or 3, especially preferably 2.

(15) When X is 2 to 5 amino acids, the 2 amino acids in the C-terminal side, that is, the 2 amino acids immediately before (PY) q, is preferably RQ, RL, RN, RM, or RT, more preferably RQ, RM, or RT.

(16) In (PY) q, Y represents an amino acid sequence composed of 1 to 4 amino acids independently selected from R, G, K, T, L, N, Q, and M, and q represents an integer of 1 to 10.

(17) More specifically, the meaning of (PY).sub.q is as follows. In PY, Y represents 1 amino acid (PY.sub.a), 2 amino acids (PY.sub.aY.sub.b), 3 amino acids (PY.sub.aY.sub.bY.sub.c), or 4 amino acids (PY.sub.aY.sub.bY.sub.cY.sub.d). Thus, (PY).sub.q means that any one or more of PY.sub.a, PY.sub.a Y.sub.b, PY.sub.a Y.sub.b Y.sub.c, and/or PY.sub.aY.sub.bY.sub.cY.sub.d continue a total of q times (wherein P represents proline). PY is preferably 3 amino acids (PY.sub.aY.sub.b) or 4 amino acids (PY.sub.a Y.sub.bY.sub.c). q is an integer of 1 to 10, preferably an integer of 2 to 10, more preferably an integer of 2 to 5, still more preferably an integer of 2 or 3, especially preferably 2.

(18) The Y's may be either the same amino acid residues or different amino acid residues selected from R, G, K, T, L, N, Q, and M. At least one of all Y's contained in the q continuous PY's (all Y's contained in the peptide tag) is preferably Q, N, L, M, or T. Two or more of the Y's are more preferably Q, N, L, M, or T. Desirably, at least one of all Y's contained in the q continuous PY's (all Y's contained in the peptide tag) is preferably Q, M, or T. Two or more of the Y's are more preferably Q, M, or T.

(19) Each PY (PY.sub.1, PY.sub.1Y.sub.2, PY.sub.1Y.sub.2Y.sub.3, or PY.sub.1Y.sub.2Y.sub.3Y.sub.4) in the q continuous PY's preferably contains one Q, N, L, M, or T, more preferably contains one Q, M, or T. In Y, the amino acid(s) other than Q, N, L, M, and T is/are preferably G ('s).

(20) For example, when Y is 2 amino acids (PY.sub.1Y.sub.2), PY is preferably PGQ, PGN, PGL, PGM, or PGT, more preferably PGQ, PGM, or PGT. When Y is 3 amino acids (PY.sub.1Y.sub.2Y.sub.3), PY is preferably PGQG, PGNG, PGLG, PGMG, or PGTG, more preferably PGQG, PGMG, or PGTG. In one preferred mode, these continue q times in an arbitrary combination in the sequence.

(21) Z represents an amino acid sequence composed of 0 to 10 amino acids independently selected

from the group consisting of R, G, S, K, T, L, N, Q, and M.
(22) When Z is 0 amino acid, the amino acid at the C-terminus of the peptide tag is P.
(23) When Z is 1 amino acid, the amino acid at the C-terminus of the peptide tag is selected from R, G, S, K, T, L, N, Q, and M.
(24) When Z is 2 to 10 amino acids, the 2 to 10 Z's may be either the same amino acid residues or different amino acid residues selected from R, G, S, K, T, L, N, Q, and M.
(25) The number of Z's is preferably 1 to 10, more preferably 1 to 5, still more preferably 2 to 5, still more preferably 2 or 3, especially preferably 2.
(26) When Z is 1 amino acid, it is preferably R or S, more preferably R.
(27) When Z is 2 to 10 amino acids, the 2 amino acids in the C-terminal side, that is, the last 2 amino acids in the peptide tag, are preferably RS.
(28) The amino acids contained in the peptide tag, that is, the amino acids contained in X, the amino acids contained in Y, and the amino acids contained in Z, preferably include three or more Q('s), N('s), L('s), M('s), and/or T('s) in total, more preferably include three or more Q('s), M('s) and/or T('s) in total. In the amino acids contained in the peptide tag, the total ratio of Q, N, L, M, and T is preferably 20 to 50%, more preferably 20 to 30%.
(29) It should be noted that the peptide tag in which the amino acids contained therein, that is, the amino acids contained in X, the amino acids contained in Y, and the amino acids contained in Z, include three or more Q('s), N('s), L('s), M('s), and/or T('s) in total, and include at least one M, is a novel peptide tag, and that a fusion protein containing the novel peptide tag, a polynucleotide encoding it, and a recombinant vector containing the polynucleotide, per se, are included within the scope of the present invention.
(30) The peptide tag used in the present invention has a length of preferably 6 to 50 amino acids, more preferably 6 to 40 amino acids, still more preferably 8 to 40 amino acids, still more preferably 10 to 30 amino acids, still more preferably 12 to 25 amino acid, especially preferably 12 to 20 amino acids.
(31) Preferred examples of the peptide tag used in the present invention include peptide tags having the following sequences:
(32) TABLE-US-00001 TABLE 1-1 RXPY.sub.1Y.sub.2Y.sub.3PY.sub.4Y.sub.5PRS No. X Y.sub.1 Y.sub.2 Y.sub.3 Y.sub.4 Y.sub.5 1-1(SEQ ID NO: 33) Q G Q G G Q 1-2(SEQ ID NO: 34) N G N G G N 1-3(SEQ ID NO: 35) M G M G G M 1-4(SEQ ID NO: 36) T G T G G T 1-5(SEQ ID NO: 37) L G L G G L 1-6(SEQ ID NO: 38) Q G Q G G N 1-7(SEQ ID NO: 39) Q G N G G Q 1-8(SEQ ID NO: 40) Q G N G G N 1-9(SEQ ID NO: 41) Q G Q G G M 1-10(SEQ ID NO: 42) Q G M G G Q 1-11(SEQ ID NO: 43) Q G M G G M 1-12(SEQ ID NO: 44) Q G Q G G T 1-13(SEQ ID NO: 45) Q G T G G Q 1-14(SEQ ID NO: 46) Q G T G G T 1-15(SEQ ID NO: 47) Q G Q G G L 1-16(SEQ ID NO: 48) Q G L G G Q 1-17(SEQ ID NO: 19) Q G L G G L 1-18(SEQ ID NO: 50) N G N G G Q 1-19(SEQ ID NO: 51) N G Q G G N 1-20(SEQ ID NO: 52) N G Q G G Q 1-21(SEQ ID NO: 53) N G N G G M 1-22(SEQ ID NO: 54) N G M G G N 1-23(SEQ ID NO: 55) N G M G G M 1-24(SEQ ID NO: 56) N G N G G T 1-25(SEQ ID NO: 57) N G T G G N 1-26(SEQ ID NO: 58) N G T G G T 1-27(SEQ ID NO: 59) N G N G G L 1-28(SEQ ID NO: 60) N G L G G N 1-29(SEQ ID NO: 61) N G L G G L 1-30(SEQ ID NO: 62) M G M G G Q 1-31(SEQ ID NO: 63) M G Q G G M 1-32(SEQ ID NO: 64) M G Q G G Q 1-33(SEQ ID NO: 65) M G M G G N 1-34(SEQ ID NO: 66) M G N G G M 1-35(SEQ ID NO: 67) M G N G G N 1-36(SEQ ID NO: 68) M G M G G T 1-37(SEQ ID NO: 69) M G T G G M 1-38(SEQ ID NO: 70) M G T G G T 1-39(SEQ ID NO: 71) M G M G G L 1-40(SEQ ID NO: 72) M G L G G M
(33) TABLE-US-00002 TABLE 1-2 RXPY.sub.1Y.sub.2Y.sub.3PY.sub.4Y.sub.5PRS No. X Y.sub.1 Y.sub.2 Y.sub.3 Y.sub.4 Y.sub.5 1-41(SEQ ID NO: 73) M G L G G L 1-42(SEQ ID NO: 74) T G T G G Q 1-43(SEQ ID NO: 75) T G Q G G T 1-44(SEQ ID NO: 76) T G Q G G Q 1-45(SEQ ID NO: 77) T G T G G N 1-46(SEQ ID NO: 78) T G N G G T 1-47(SEQ ID NO: 79) T G N G G N 1-48(SEQ ID NO: 80) T G T G G M 1-49(SEQ ID NO: 81) T G M G G T 1-50(SEQ ID NO: 82) T G

M G G M 1-51(SEQ ID NO: 83) T G T G G L 1-52(SEQ ID NO: 84) T G L G G T 1-53(SEQ ID NO: 85) T G L G G L 1-54(SEQ ID NO: 86) L G L G G Q 1-55(SEQ ID NO: 87) L G Q G G L 1-56(SEQ ID NO: 88) L G Q G G Q 1-57(SEQ ID NO: 89) L G L G G N 1-58(SEQ ID NO: 90) L G N G G L 1-59(SEQ ID NO: 91) L G N G G N 1-60(SEQ ID NO: 92) L G L G G M 1-61(SEQ ID NO: 93) L G M G G L 1-62(SEQ ID NO: 94) L G M G G M 1-63(SEQ ID NO: 95) L G L G G T 1-64(SEQ ID NO: 96) L G T G G L 1-65(SEQ ID NO: 97) L G T G G T 1-66(SEQ ID NO: 98) Q Q Q Q Q Q 1-67(SEQ ID NO: 99) N N N N N N 1-68(SEQ ID NO: 100) M M M M M M 1-69(SEQ ID NO: 101) T T T T T T 1-70(SEQ ID NO: 102) L L L L L L

(34) TABLE-US-00003 TABLE 1-3 RXPY.sub.1Y.sub.2Y.sub.3PY.sub.4Y.sub.5GRS No. X Y.sub.1 Y.sub.2 Y.sub.3 Y.sub.4 Y.sub.5 2-1(SEQ ID NO: 103) Q G Q G G Q 2-2(SEQ ID NO: 104) N G N G G N 2-3(SEQ ID NO: 105) M G M G G M 2-4(SEQ ID NO: 106) T G T G G T 2-5(SEQ ID NO: 107) L G L G G L 2-6(SEQ ID NO: 108) Q G Q G G N 2-7(SEQ ID NO: 109) Q G N G G Q 2-8(SEQ ID NO: 110) Q G N G G N 2-9(SEQ ID NO: 111) Q G Q G G M 2-10(SEQ ID NO: 112) Q G M G G Q 2-11(SEQ ID NO: 113) Q G M G G M 2-12(SEQ ID NO: 114) Q G Q G G T 2-13(SEQ ID NO: 115) Q G T G G Q 2-14(SEQ ID NO: 116) Q G T G G T 2-15(SEQ ID NO: 117) Q G Q G G L 2-16(SEQ ID NO: 118) Q G L G G Q 2-17(SEQ ID NO: 119) Q G L G G L 2-18(SEQ ID NO: 120) N G N G G Q 2-19(SEQ ID NO: 121) N G Q G G N 2-20(SEQ ID NO: 122) N G Q G G Q 2-21(SEQ ID NO: 123) N G N G G M 2-22(SEQ ID NO: 124) N G M G G N 2-23(SEQ ID NO: 125) N G M G G M 2-24(SEQ ID NO: 126) N G N G G T 2-25(SEQ ID NO: 127) N G T G G N 2-26(SEQ ID NO: 128) N G T G G T 2-27(SEQ ID NO: 129) N G N G G L 2-28(SEQ ID NO: 130) N G L G G N 2-29(SEQ ID NO: 131) N G L G G L 2-30(SEQ ID NO: 132) M G M G G Q 2-31(SEQ ID NO: 133) M G Q G G M 2-32(SEQ ID NO: 134) M G Q G G Q 2-33(SEQ ID NO: 135) M G M G G N 2-34(SEQ ID NO: 136) M G N G G M 2-35(SEQ ID NO: 137) M G N G G N 2-36(SEQ ID NO: 138) M G M G G T 2-37(SEQ ID NO: 139) M G T G G M 2-38(SEQ ID NO: 140) M G T G G T 2-39(SEQ ID NO: 141) M G M G G L 2-40(SEQ ID NO: 142) M G L G G M

(35) TABLE-US-00004 TABLE 1-4 RXPY.sub.1Y.sub.2Y.sub.3PY.sub.4Y.sub.5GRS No. X Y.sub.1 Y.sub.2 Y.sub.3 Y.sub.4 Y.sub.5 2-41(SEQ ID NO: 143) M G L G G L 2-42(SEQ ID NO: 144) T G T G G Q 2-43(SEQ ID NO: 145) T G Q G G T 2-44(SEQ ID NO: 146) T G Q G G Q 2-45(SEQ ID NO: 147) T G T G G N 2-46(SEQ ID NO: 148) T G N G G T 2-47(SEQ ID NO: 149) T G N G G N 2-48(SEQ ID NO: 150) T G T G G M 2-49(SEQ ID NO: 151) T G M G G T 2-50(SEQ ID NO: 152) T G M G G M 2-51(SEQ ID NO: 153) T G T G G L 2-52(SEQ ID NO: 154) T G L G G T 2-53(SEQ ID NO: 155) T G L G G L 2-54(SEQ ID NO: 156) L G L G G Q 2-55(SEQ ID NO: 157) L G Q G G L 2-56(SEQ ID NO: 158) L G Q G G Q 2-57(SEQ ID NO: 159) L G L G G N 2-58(SEQ ID NO: 160) L G N G G L 2-59(SEQ ID NO: 161) L G N G G N 2-60(SEQ ID NO: 162) L G L G G M 2-61(SEQ ID NO: 163) L G M G G L 2-62(SEQ ID NO: 164) L G M G G M 2-63(SEQ ID NO: 165) L G L G G T 2-64(SEQ ID NO: 166) L G T G G L 2-65(SEQ ID NO: 167) L G T G G T 2-66(SEQ ID NO: 168) Q Q Q Q Q Q 2-67(SEQ ID NO: 169) N N N N N N 2-68(SEQ ID NO: 170) M M M M M M 2-69(SEQ ID NO: 171) T T T T T T 2-70(SEQ ID NO: 172) L L L L L L

(36) TABLE-US-00005 TABLE 1-5 RXPY.sub.1Y.sub.2PY.sub.3Y.sub.4PY.sub.5RS No. X Y.sub.1 Y.sub.2 Y.sub.3 Y.sub.4 Y.sub.5 3-1(SEQ ID NO: 173) Q G Q G Q G 3-2(SEQ ID NO: 174) N G N G N G 3-3(SEQ ID NO: 175) M G M G M G 3-4(SEQ ID NO: 176) T G T G T G 3-5(SEQ ID NO: 177) L G L G L G 3-6(SEQ ID NO: 178) Q G Q G N G 3-7(SEQ ID NO: 179) Q G N G Q G 3-8(SEQ ID NO: 180) Q G N G N G 3-9(SEQ ID NO: 181) Q G Q G M G 3-10(SEQ ID NO: 182) Q G M G Q G 3-11(SEQ ID NO: 183) Q G M G M G 3-12(SEQ ID NO: 184) Q G Q G T G 3-13(SEQ ID NO: 185) Q G T G Q G 3-14(SEQ ID NO: 186) Q G T G T G 3-15(SEQ ID NO: 187) Q G Q G L G 3-16(SEQ ID NO: 188) Q G L G Q G 3-17(SEQ ID NO: 189) Q G L G L G 3-18(SEQ ID NO: 190) N G N G Q G 3-19(SEQ ID NO: 191) N G Q G N G 3-20(SEQ ID NO: 192) N G Q G Q G 3-21(SEQ ID NO: 193) N G N G M G 3-22(SEQ ID NO: 194) N G M G N G 3-

23(SEQ ID NO: 195) N G M G M G 3-24(SEQ ID NO: 196) N G N G T G 3-25(SEQ ID NO: 197) N G T G N G 3-26(SEQ ID NO: 198) N G T G T G 3-27(SEQ ID NO: 199) N G N G L G 3-28(SEQ ID NO: 200) N G L G N G 3-29(SEQ ID NO: 201) N G L G L G 3-30(SEQ ID NO: 202) M G M G Q G 3-31(SEQ ID NO: 203) M G Q G M G 3-32(SEQ ID NO: 204) M G Q G Q G 3-33(SEQ ID NO: 205) M G M G N G 3-34(SEQ ID NO: 206) M G N G M G 3-35(SEQ ID NO: 207) M G N G N G 3-36(SEQ ID NO: 208) M G M G T G 3-37(SEQ ID NO: 209) M G T G M G 3-38(SEQ ID NO: 210) M G T G T G 3-39(SEQ ID NO: 211) M G M G L G 3-40(SEQ ID NO: 212) M G L G M G

(37) TABLE-US-00006 TABLE 1-6 RXPY.sub.1Y.sub.2PY.sub.3Y.sub.4PY.sub.5RS No. X Y.sub.1 Y.sub.2 Y.sub.3 Y.sub.4 Y.sub.5 3-41(SEQ ID NO: 213) M G L G L G 3-42(SEQ ID NO: 214) T G T G Q G 3-43(SEQ ID NO: 215) T G Q G T G 3-44(SEQ ID NO: 216) T G Q G Q G 3-45(SEQ ID NO: 217) T G T G N G 3-46(SEQ ID NO: 218) T G N G T G 3-47(SEQ ID NO: 219) T G N G N G 3-48(SEQ ID NO: 220) T G T G M G 3-49(SEQ ID NO: 221) T G M G T G 3-50(SEQ ID NO: 222) T G M G M G 3-51(SEQ ID NO: 223) T G T G L G 3-52(SEQ ID NO: 224) T G L G T G 3-53(SEQ ID NO: 225) T G L G L G 3-54(SEQ ID NO: 226) L G L G Q G 3-55(SEQ ID NO: 227) L G Q G L G 3-56(SEQ ID NO: 228) L G Q G Q G 3-57(SEQ ID NO: 229) L G L G N G 3-58(SEQ ID NO: 230) L G N G L G 3-59(SEQ ID NO: 231) L G N G N G 3-60(SEQ ID NO: 232) L G L G M G 3-61(SEQ ID NO: 233) L G M G L G 3-62(SEQ ID NO: 234) L G M G M G 3-63(SEQ ID NO: 235) L G L G T G 3-64(SEQ ID NO: 236) L G T G L G 3-65(SEQ ID NO: 237) L G T G T G 3-66(SEQ ID NO: 238) Q Q Q Q Q Q 3-67(SEQ ID NO: 239) N N N N N N 3-68(SEQ ID NO: 240) M M M M. M M 3-69(SEQ ID NO: 241) T T T T T T 3-70(SEQ ID NO: 242) L L L L L L

(38) In the present invention, the fusion protein is a protein in which the peptide tag is linked to a protein of interest. The peptide tag may be linked to the N-terminus of the protein of interest; the peptide tag may be linked to the C-terminus of the protein of interest; or the peptide tag may be linked to each of the N-terminus and C-terminus of the protein of interest. The peptide tag(s) may be directly linked to the N-terminus and/or C-terminus of the protein of interest, or may be linked thereto through a sequence(s) of one to several amino acids (for example, 1 to 5 amino acids). The sequence of one to several amino acids may be an arbitrary sequence as long as the sequence does not adversely affect the function and the expression level of the protein of interest. In cases where the sequence is a protease recognition sequence, the peptide tag may be cleaved off from the protein of interest after the expression and purification. Examples of the protease recognition sequence include a factor Xa recognition sequence. The fusion protein may also include another tag sequence required for detection or purification, such as a His tag, HN tag, or FLAG tag.

(39) The type of the protein of interest contained in the fusion protein is not limited. The protein of interest is preferably a protein to be used for medical application, diagnosis, or material production. Examples of the protein of interest include growth factors, hormones, cytokines, blood proteins, enzymes, antigens, antibodies, transcription factors, receptors, fluorescent proteins, and partial peptides thereof.

(40) Examples of the enzymes include lipase, protease, steroid-synthesizing enzymes, kinase, phosphatase, xylanase, esterase, methylase, demethylase, oxidase, reductase, cellulase, aromatase, collagenase, transglutaminase, glycosidase, and chitinase.

(41) Examples of the growth factors include epidermal growth factor (EGF), insulin-like growth factor (IGF), transforming growth factor (TGF), nerve growth factor (NGF), brain-derived neurotrophic factor (BDNF), vascular endothelial growth factor (VEGF), granulocyte colony-stimulating factor (G-CSF), granulocyte-macrophage colony-stimulating factor (GM-CSF), platelet-derived growth factor (PDGF), erythropoietin (EPO), thrombopoietin (TPO), fibroblast growth factor (FGF), and hepatocyte growth factor (HGF).

(42) Examples of the hormones include insulin, glucagon, somatostatin, growth hormone, parathyroid hormone, prolactin, leptin, and calcitonin.

(43) Examples of the cytokines include interleukins, interferons (IFNα, IFNß, IFNγ), and tumor

necrosis factor (TNF).
(44) Examples of the blood proteins include thrombin, serum albumin, factor VII, factor VIII, factor IX, factor X, and tissue plasminogen activator.
(45) Examples of the antibodies include complete antibodies, Fab, F(ab′), F(ab′).sub.2, Fc, Fc fusion proteins, heavy chain (H-chain), light chain (L-chain), single-chain Fv (scFv), sc(Fv).sub.2, disulfide-linked Fv (sdFv), and diabodies.
(46) For use as vaccines, the antigen proteins are not limited as long as the immune response can be induced therewith. The antigen proteins may be appropriately selected depending on the expected target of the immune response. Examples of the antigen proteins include proteins derived from pathogenic bacteria and proteins derived from pathogenic viruses.
(47) To the fusion protein, a secretion signal peptide that functions in a host cell may be added for secretory production. Examples of the secretion signal peptide include: invertase secretion signal, P3 secretion signal, and α-factor secretion signal in cases where yeast is used as the host; PelB secretion signal in cases where *E. coli* is used as the host; and P2 secretion signal and P22 secretion signal in cases where *Brevibacillus* is used as the host. When a plant is used as the host, examples of the secretion signal include those derived from tobacco (*Nicotiana tabacum*), *Arabidopsis thaliana*, strawberry (*Fragaria* x *ananassa*), lettuce (*Lactuca sativa*), or the like.
(48) For allowing expression of the fusion protein in a particular cellular compartment, a transport signal peptide such as an endoplasmic reticulum retention signal peptide or a vacuole transport signal peptide may be added thereto.
(49) By introducing a polynucleotide encoding the fusion protein into an expression system, the fusion protein can be produced by genetic engineering.
(50) In the present invention, the polynucleotide means a substance having a base sequence carrying genetic information encoding the fusion protein. Examples of the polynucleotide include DNA and RNA. More specifically, the polynucleotide encoding the fusion protein contains a polynucleotide encoding the protein of interest and a polynucleotide encoding the peptide tag, wherein the polynucleotide encoding the protein of interest is linked to the polynucleotide encoding the peptide tag in the same reading frame.
(51) The polynucleotide encoding the protein of interest may be obtained by, for example, a common genetic engineering method based on a known base sequence.
(52) Preferably, in the polynucleotide encoding the fusion protein, a codon(s) corresponding to an amino acid(s) constituting the fusion protein is/are modified as appropriate such that the translation level of the fusion protein increases depending on the host cell in which the protein is to be produced. For the method of the codon modification, one may refer to, for example, the method of Kang et al. (Protein Expr Purif. 2004 November; 38 (1): 129-35). Examples of the method also include methods in which codons frequently used in the host cell are selected, methods in which codons with high GC contents are selected, and methods in which codons frequently used in housekeeping genes of the host cell are selected.
(53) For improving expression in the host cell, the polynucleotide encoding the fusion protein may contain an enhancer sequence or the like that functions in the host cell.
(54) The polynucleotide encoding the fusion protein may be prepared by a common genetic engineering method. For example, a DNA encoding the peptide tag, a DNA encoding the protein of interest, and the like may be linked to each other using PCR, DNA ligase, and/or the like, to construct the polynucleotide.
(55) The polynucleotide encoding the fusion protein may be introduced as it is into the expression system. The polynucleotide is, however, preferably introduced into the expression system as a recombinant vector containing the polynucleotide.
(56) The recombinant vector may be a vector in which the polynucleotide encoding the fusion protein is inserted such that the expression is possible in the host cell to which the vector is to be introduced. The vector is not limited as long as it can replicate in the host cell. Examples of the

vector include plasmid DNAs and viral DNAs. The vector preferably contains a selection marker such as a drug resistance gene. Specific examples of the plasmid vectors include pTrcHis2 vector, pUC119, pBR322, pBluescript II KS+, pYES2, pAUR123, pQE-Tri, pET, pGEM-3Z, pGEX, pMAL, pRI909, pRI910, pBI221, pBI121, pNCMO2, pBI101, pIG121Hm, pTrc99A, pKK223, pA1-11, pXT1, pRc/CMV, pRc/RSV, pcDNA I/Neo, p3×FLAG-CMV-14, pCAT3, pcDNA3.1, and pCMV.
(57) The promoter used in the vector may be appropriately selected depending on the host cell to which the vector is to be introduced. In cases of expression in yeast, examples of the promoter that may be used include the GAL1 promoter, PGK1 promoter, TEF1 promoter, ADH1 promoter, TPI1 promoter, and PYK1 promoter. In cases of expression in mammalian cells, examples of the promoter that may be used include the CMV promoter, SV40 promoter, and EF1α promoter. In cases of expression in a plant, examples of the promoter that may be used include the cauliflower mosaic virus 35S promoter, rice actin promoter, maize ubiquitin promoter, and lettuce ubiquitin promoter. In cases of expression in *E. coli*, examples of the promoter include the T7 promoter. In cases of expression in *Brevibacillus*, examples of the promoter include the P2 promoter and the P22 promoter. The promoter may be an inducible promoter. Examples of the inducible promoter that may be used include lac, tac, and trc, which are inducible with IPTG; trp, which is inducible with IAA; ara, which is inducible with L-arabinose; Pzt-1, which is inducible with tetracycline; the Pz promoter, which is inducible by heat (42° C.); and the promoter of the cspA gene, which is a cold shock gene.
(58) When necessary, a terminator sequence may also be included depending on the host cell.
(59) The recombinant vector may be prepared by, for example, cleaving a DNA encoding the fusion protein with an appropriate restriction enzyme, or adding a restriction site thereto by PCR, and then inserting the resulting DNA into a restriction site or a multicloning site in a vector.
(60) In the present invention, the polynucleotide encoding the fusion protein, or the recombinant vector containing it, may be introduced into an expression system, and may then be allowed to express the fusion protein in the expression system, to enable efficient accumulation of the fusion protein in a soluble fraction.
(61) The expression system herein means a system comprising translation factors required for protein expression, such as ribosomes, tRNAs, and amino acids, which system is capable of producing a fusion protein from the polynucleotide encoding the fusion protein, or from a recombinant vector containing it. Examples of the expression system include: prokaryotic cells, such as bacterial cells, including *E. coli*, *Bacillus* (bacteria belonging to the genus *Bacillus*), bacteria belonging to the genus *Brevibacillus*, actinomycetes, corynebacteria, and cyanobacteria; and eukaryotic cells, such as yeast cells including baker's yeast, *Pichia* yeast, and fission yeast, aspergilli, insect cells, mammalian cells, and plant cells. Examples of the expression system also include: cell-free protein expression systems derived from prokaryotic cells such as *E. coli*; and cell-free protein expression systems derived from eukaryotic cells such as reticulocytes, wheat germ, or insect cell extracts.
(62) The expression system is preferably transformed cells obtained by transformation of an expression system with the polynucleotide encoding the fusion protein, or with a recombinant vector containing it. The transformed cells may be prepared by introducing the polynucleotide or recombinant vector encoding the fusion protein into host cells, using a common genetic engineering method. Examples of the method that may be used include the electroporation method (Tada, et al., 1990, Theor. Appl. Genet, 80:475), the protoplast method (Gene, 39, 281-286 (1985)), the polyethylene glycol method (Lazzeri, et al., 1991, Theor. Appl. Genet. 81:437), introduction methods utilizing *Agrobacterium* (Hood, et al., 1993, Transgenic, Res. 2:218, Hiei, et al., 1994 Plant J. 6:271), the particle gun method (Sanford, et al., 1987, J. Part. Sci. tech. 5:27), and the polycation method (Ohtsuki, et al., FEBS Lett. 1998 May 29; 428 (3): 235-40). The gene expression may be transient expression, or may be stable expression based on incorporation into

the chromosome. The transformant may be selected utilizing a selection marker such as a drug resistance gene.
(63) In the expression systems such as the transformed cells and the cell-free protein expression systems, the fusion protein is accumulated in a soluble fraction. Conditions for the protein expression, such as the medium, temperature, and time, may be appropriately selected in accordance with the type of the expression system.
(64) The soluble fraction means a fraction containing the medium outside the cells, the intracellular liquid fraction which contains neither nuclei nor organelles, and the like. The soluble fraction thus means a fraction from which the fusion protein can be collected as a solution.
(65) The fusion protein is preferably produced as a nondenatured protein. The nondenatured protein herein means a protein which maintains its higher-order structure, and which has not undergone changes such as activity loss or insolubilization, or has undergone such changes only to a small extent.
(66) By collecting the soluble fraction, a solution containing the nondenatured protein can be obtained.
(67) The solution containing the nondenatured protein herein means a state where the nondenatured protein is solubilized in water in the cells and outside the cells. It is a state where, in the cells, the expressed protein is located, without aggregation, in the cytosol or in its proper positions in the cells such that the protein can easily exert its original function or activity, and where, outside the cells, the protein of interest is solubilized, without aggregation, in a solvent composed mainly of water, such that the solubilized protein has its original function or activity.
(68) The fusion protein accumulated in the soluble fraction may be used as it is for enzymatic reaction or the like. Alternatively, the fusion protein may be separated and purified according to a method well known to those skilled in the art. For example, the separation and purification may be carried out by an appropriate known method such as salting-out, ethanol precipitation, ultrafiltration, gel filtration chromatography, ion-exchange column chromatography, affinity chromatography, high/medium-pressure liquid chromatography, reversed-phase chromatography, or hydrophobic chromatography, or by combination of any of these.
EXAMPLES
(69) Examples of the present invention are described below, but the present invention is not limited to the Examples.
(70) (1) Construction of Gene Expression Plasmid Encoding Peptide-Tagged Esterase for *Brevibacillus*, and Transformation Therewith
(71) An artificial synthetic DNA (SEQ ID NO:26) encoding esterase derived from *Bacillus subtilis* (EstA, SEQ ID NO:25) was inserted into the EcoRV recognition site of the pUC19-modified plasmid pUCFa (Fasmac), thereby plasmid 1 was obtained.
(72) As a plasmid for expression in *Brevibacillus*, pNCMO2 (Takara Bio Inc.) was used (plasmid 2).
(73) By the following procedure, plasmids for expressing, in *Brevibacillus*,
(74) fusion proteins in which various peptide tags are added to the N-terminus of esterase, and in which a 6×His tag for detection and purification is added to the C-terminus, and
(75) fusion proteins in which a 6×His tag for detection and purification is added to the N-terminus of esterase, and in which various peptide tags are added to the C-terminus,
(76) were constructed (FIGS. **1** and **2**).
(77) TABLE-US-00007 TABLE 2 Amino acid sequence of each peptide tag Position of Amino acid No. Tag addition sequence Comparative — Tag(-)-N — — Example 1 Comparative — PG12-N N terminus RSPGSGPGSPRS Example 2 (SEQ ID NO: 1) Comparative — PX12-20-N N terminus RKPGKGPGKPRS Example 3 (SEQ ID NO: 4) Example 1 1-1 PX12-32-N N terminus RQPGQGPGQPRS (SEQ ID NO: 7) Example 2 1-2 PX12-33-N N terminus RNPGNGPGNPRS (SEQ ID NO: 10) Example

3 1-3 PX12-34-N N terminus RMPGMGPGMPRS (SEQ ID NO: 12) Example 4 1-4 PX12-35-N N terminus RTPGTGPGTPRS (SEQ ID NO: 15) Example 5 1-5 PX12-36-N N terminus RLPGLGPGLPRS (SEQ ID NO: 17) Example 6 1-66 PX12-90-N N terminus RQPQQQPQQPRS (SEQ ID NO: 23) Example 7 2-1 PX12-89-N N-terminus RQPGQGPGQGRS (SEQ ID NO: 21) Example 8 3-3 PX12-83-N N terminus RMPGMPGMPGRS (SEQ ID NO: 19) Comparative — Tag(-)-C — — Example 4 Comparative — PG12-C C terminus RSPGSGPGSPRS Example 5 (SEQ ID NO: 1) Comparative — PX12-20-C C terminus RKPGKGPGKPRS Example 6 (SEQ ID NO: 4) Example 9 1-1 PX12-32-C C terminus RQPGQGPGQPRS (SEQ ID NO: 7) Exmample 10 1-3 PX12-34-C C terminus RMPGMGPGMPRS (SEQ ID NO: 10)

(78) First, for the addition of the various peptide tags to the N- or C-terminus of esterase, PCR was carried out using the combinations of a template plasmid, a forward primer, and a reverse primer shown in Table 3.

(79) TABLE-US-00008 TABLE 3 Template plasmid and primers used for PCR amplification of BsestA gene added with peptide tag-coding sequence PCR amplified Added peptide tag sequence Template fragment N terminus C terminus Plasmid Forward primer Reverse primer Tag-(—)-N — — Plasmid1 mCWSP-AD-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 29) (SEQ ID NO: 32) Tag-(—)-C — — Plasmid1 mCWSP-AD-6xHis-BsestAF BsestAR-stopXt (SEQ ID NO: 31) (SEQ ID NO: 30) BsestA-PG12-N SEQ ID NO: 1 — Plasmid1 mCWSP-AD-PG12-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 2) (SEQ ID NO: 32) BsestA-PG12-C — SEQ ID NO: 1 Plasmid1 mCWSP-AD-6xHis-BsestAF BsestAR-PG12-stopXt (SEQ ID NO: 31) (SEQ ID NO: 3) BsestA-PX12-20-N SEQ ID NO: 4 — Plasmid1 mCWSP-AD-PX12-20-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 5) (SEQ ID NO: 32) BsestA-PX12-20-C — SEQ ID NO: 4 Plasmid1 mCWSP-AD-6xHis-BsestAF BsestAR-PX12-stopXt (SEQ ID NO: 31) (SEQ ID NO: 6) BsestA-PX12-32-N SEQ ID NO: 7 — Plasmid1 mCWSP-AD-PX12-32-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 8) (SEQ ID NO: 32) BsestA-PX12-32-C — SEQ ID NO: 7 Plasmid1 mCWSP-AD-6xHis-BsestAF BsestAR-PX12-stopXt (SEQ ID NO: 31) (SEQ ID NO: 9) BsestA-PX12-33-N SEQ ID NO: 10 — Plasmid1 mCWSP-AD-PX12-33-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 11) (SEQ ID NO: 32) BsestA-PX12-34-N SEQ ID NO: 12 — Plasmid1 mCWSP-AD-PX12-34-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 13) (SEQ ID NO: 32) BsestA-PX12-34-C — SEQ ID NO: 12 Plasmid1 mCWSP-AD-6xHis-BsestAF BsestAR-PX12-stopXt (SEQ ID NO: 31) (SEQ ID NO: 14) BsestA-PX12-35-N SEQ ID NO: 15 — Plasmid1 mCWSP-AD-PX12-35-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 16) (SEQ ID NO: 32) BsestA-PX12-36-N SEQ ID NO: 17 — Plasmid1 mCWSP-AD-PX12-36-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 18) (SEQ ID NO: 32) BsestA-PX12-83-N SEQ ID NO: 19 — Plasmid1 mCWSP-AD-PX12-83-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 20) (SEQ ID NO: 32) BsestA-PX12-89-N SEQ ID NO: 21 — Plasmid1 mCWSP-AD-PX12-89-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 22) (SEQ ID NO: 32) BsestA-PX12-90-N SEQ ID NO: 23 — Plasmid1 mCWSP-AD-PX12-90-BsestAF BsestAR-6xHis-stopXt (SEQ ID NO: 24) (SEQ ID NO: 32)

(80) To the 5′-end of each primer, a sequence homologous to plasmid 2 was added. In designing of the forward primer, the 2 amino acid residues AD were added such that they follow a signal peptide. For the PCR, KOD-PLUS-Ver. 2 (Toyobo Co., Ltd.) was used. A reaction liquid in an amount of 50 μl was prepared such that it contained 2 pg/μl template plasmid, 0.3 μM forward primer, 0.3 μM reverse primer, 0.2 mM dNTPs, 1× Buffer for KOD-Plus-Ver. 2, 1.5 mM MgSO.sub.4, and 0.02 U/μl KOD-PLUS-Ver. 2. The reaction liquid was heated at 94° C. for 5 minutes, and this was followed by 30 cycles of treatment each composed of heating at 98° C. for 10 seconds, at 60° C. for 30 seconds, and then at 68° C. for 40 seconds. Finally, the reaction liquid was heated at 68° C. for 5 minutes.

(81) The resulting amplification fragment was purified with a QIAquick PCR Purification Kit (Qiagen).

(82) Plasmid 2 was digested with NcoI and HindIII, and then separated by electrophoresis using 0.8% SeaKem GTG Agarose, followed by extraction from the gel using a QIAquick Gel Extraction Kit (Qiagen).
(83) About 50 ng of the extracted plasmid 2 was mixed with 2 μl of the purified PCR product, and the liquid volume was adjusted to 5 μl. The resulting mixture was mixed with 5 μl of 2× Enzyme Mix included in a Gene Art Seamless PLUS Cloning and Assembly Kit (Thermo Fisher Scientific), and then left to stand at room temperature for 30 minutes, followed by being left to stand on ice for 5 minutes.
(84) With competent cells DH5-α, 5 μl of the reaction liquid was mixed, and the resulting mixture was left to stand on ice for 30 minutes. The mixture was then warmed at 42° C. for 45 seconds, and left to stand on ice for 2 minutes, followed by addition of 250 μl of SOC thereto and shaking at 37° C. at 200 rpm for 1 hour. Subsequently, 50 μl of the shaken product was applied to 2×YT agar medium supplemented with 100 mg/l ampicillin, and static culture was carried out at 37° C. overnight, to obtain transformed colonies. A colony was transferred to 2×YT liquid medium supplemented with 100 mg/l ampicillin, and shake culture was carried out at 37° C. at 200 rpm overnight, followed by extraction of the plasmid for gene expression. After confirmation of the base sequence, the plasmid was used for transformation of *Brevibacillus*.
(85) Competent cells of the *Brevibacillus choshinensis* SP3 strain were left to stand on a heat block at 37° C. for 30 seconds to allow rapid thawing, and then centrifuged (12,000 rpm, room temperature, 1 minute). After removing the supernatant, the whole amount of a mixture of 1 μl of the above plasmid solution for gene expression and 50 μl of Solution A was added to the cells, and the pellet of the competent cells was completely suspended by vortexing, followed by leaving the resulting suspension to stand for 5 minutes. After addition of 150 μl of Solution B, the suspension was mixed by vortexing for 10 seconds, and then centrifugation was carried out (5000 rpm, room temperature, 5 minutes), followed by removing the supernatant. After carrying out centrifugation (5000 rpm, room temperature, 30 seconds) again, the supernatant was completely removed. To the resulting pellet, 1 ml of MT medium as added, and the pellet was completely suspended using a micropipette, followed by shake culture at 37° C. at 200 rpm for 1 hour. The culture liquid was plated on an MTNm plate, and static culture was carried out at 37° C. overnight, to obtain transformed *Brevibacillus*.
(86) (2) Protein Expression Culture of *Brevibacillus*
(87) A single colony of the transformed *Brevibacillus* was smeared on an MTNm plate, and left to stand at 30° C. overnight to perform culture. Subsequently, bacterial cells were scraped with a 1-μl sterile disposable loop from the plate medium after the culture, and then inoculated into 5 ml of TMNm medium preliminarily dispensed in a sterile 50-ml polystyrene tube. Shake culture was then carried out at 30° C. at 120 rpm for 48 hours. After completion of the culture, the culture liquid containing bacterial cells was sampled.
(88) The culture liquid containing bacterial cells was aliquoted in 100-μl volumes into 1.5-ml Eppendorf tubes, and centrifuged (8,000× g, 4° C., 10 minutes) to separate the bacterial cells from the culture supernatant, followed by storing 50 μl of the culture supernatant and the whole amount of bacterial cells at −80° C.
(89) (3) Extraction of Protein from *Brevibacillus*
(90) To 50 μl of the cryopreserved culture supernatant, 50 μl of 2×sample buffer (EZ Apply, manufactured by ATTO) was added, and the resulting mixture was stirred using a vortex mixer, followed by heating the mixture in boiling water for 10 minutes to perform SDS treatment.
(91) On the other hand, the bacterial cells were homogenized in 100 μl of xTractor Buffer (Takara Bio Inc.) containing 0.01% SEM-nuclease (Wako), and then centrifuged at 10,000×g for 10 minutes at 4° C. Thereafter, 50 μl of 2× sample buffer was added to 50 μl of the supernatant fraction (intracellular soluble fraction), and SDS treatment was carried out by the same procedure.
(92) (4) Western Analysis

(93) As a standard substance for protein quantification, an esterase (EstA) preparation was used. By 2-fold serial dilution of the preparation with a sample buffer, a dilution series to be used as standards was prepared.
(94) For electrophoresis (SDS-PAGE) of protein, an electrophoresis tank (Criterion cell, BIO RAD) and Criterion TGX-gel (BIO RAD) were used. In the electrophoresis tank, an electrophoresis buffer (Tris/Glycine/SDS Buffer, BIO RAD) was placed, and 4 μl of the SDS-treated sample was applied to each well, followed by performing electrophoresis at a constant voltage of 200 V for 40 minutes.
(95) The gel after the electrophoresis was subjected to blotting by Trans-Blot Turbo (BIO RAD) using a Trans-Blot Transfer Pack (BIO RAD).
(96) The membrane after the blotting was immersed in a blocking solution (TBS system, pH 7.2; Nacalai Tesque, Inc.), and shaken at room temperature for 1 hour or left to stand at 4° C. for 16 hours. Thereafter, the membrane was washed by three times of shaking in TBS-T (137 mM sodium chloride, 2.68 mM potassium chloride, 1% polyoxyethylene sorbitan monolaurate, 25 mM Tris-HCl, pH 7.4) at room temperature for 5 minutes.
(97) For detection of esterase, an antiserum Mouse-monoclonal Anti-6×His antibody (Abcam) diluted 6,000-fold with TBS-T was used. The membrane was immersed in the dilution, and shaken at room temperature for 2 hours to allow antigen-antibody reaction, followed by three times of washing by shaking in TBS-T at room temperature for 5 minutes. As a secondary antibody, an Anti-Mouse IgG, AP-linked Antibody (Cell Signaling TECHNOLOGY) diluted 3000-fold with TBS-T was used.
(98) The membrane was immersed in the dilution, and shaken at room temperature for 1 hour to allow antigen-antibody reaction, followed by three times of washing by shaking in TBS-T at room temperature for 5 minutes. Chromogenic reaction with alkaline phosphatase was carried out by immersing the membrane in a coloring solution (0.1 M sodium chloride, 5 mM magnesium chloride, 0.33 mg/ml nitroblue tetrazolium, 0.33 mg/ml 5-bromo-4-chloro-3-indolyl-phosphate, 0.1 M Tris-HCl, pH 9.5), and shaking the membrane at room temperature for 15 minutes. The membrane was washed with distilled water, and then dried at normal temperature.
(99) From the membrane after the coloring, an image was obtained using a scanner (PM-A900, Epson) at a resolution of 600 dpi, and esterase in the culture supernatant and in the bacterial cell-solubilized fraction was quantified using image analysis software (CS Analyzer ver. 3.0, Atto Corporation). The total amount of the enzyme in the culture supernatant and in the bacterial cell-solubilized fraction was regarded as the total amount of solubilized esterase, and subjected to comparison.
(100) (5) Providing of Cultured Sample for Enzyme Purification
(101) The *Brevibacillus* strain constructed for esterase expression, stored at −80° C., was smeared on an MTNm plate using a sterile loop, and cultured at 37° C. for 16 to 20 hours. Subsequently, 5 ml of TMNm was dispensed into a 50-ml polypropylene conical tube, and the plate sample was inoculated thereto with a sterile loop. After performing shake culture at 30° C. at 120 rpm for 16 to 20 hours, the whole amount of the culture was added to a 500-ml baffled Erlenmeyer flask containing 150 ml of TMNm medium dispensed therein, and shake culture was carried out at 120 rpm at 30° C. for 48 hours. Into a 50-ml polypropylene conical tube, 50 ml of the culture was taken, and centrifugation was carried out at 8,000×g for 10 minutes at 4° C. After precipitation of the bacterial cells, 40 ml of the supernatant was transferred to another tube, and the remaining supernatant was completely removed with a pipet. The tubes each containing the supernatant or the bacterial cells (precipitate) were frozen with liquid nitrogen, and then stored at −80° C.
(102) (6) Purification of His-Tag Fusion Enzyme Protein from Supernatant Fraction
(103) After thawing 40 ml of the supernatant fraction cryopreserved at −80° C., 13.3 ml of 0.4 M sodium phosphate buffer (pH 7.4) containing 1.2 M NaCl and 0.04 M imidazole was added thereto and mixed therewith. An EcoPack column (BIO RAD) was packed with 5 ml of TALON His-tag fusion protein purification resin (Clontech), and then equilibrated with 25 ml of an equilibration

buffer (0.1 M sodium phosphate buffer (pH 7.4) containing 0.3 M NaCl and 0.01 M imidazole). About 10 ml of the supernatant sample prepared with phosphate buffer was charged into the column, and the resin was suspended therein. Using the remaining sample, the whole amount of the resin was washed out into a 100-ml polypropylene centrifuge tube (manufactured by IWAKI). The centrifuge tube containing the sample was placed on a rotator, and the His-tag fusion enzyme protein was allowed to adsorb to the resin for 2 hours in a low-temperature chamber (4° C.).

(104) Subsequently, the resin to which the sample was adsorbed, suspended in a sample buffer, was transferred to an empty column, and only the buffer was eluted therefrom. Further, the resin was washed by allowing 50 ml of the equilibration buffer to flow through the column.

(105) Thereafter, the His-tag fusion enzyme protein adsorbed to the resin was eluted using 20 ml of an elution buffer (0.1 M sodium phosphate buffer (pH 7.4) containing 0.3 M NaCl and 0.15 M imidazole). The elution was carried out by repeating 10 times an operation of taking a 2-ml fraction into a 2-ml Eppendorf tube ice-cooled on an aluminum block, to obtain 10 tubes containing separate fractions. Twenty microliters of each fraction was taken into a 1.5-ml Eppendorf tube, and 20 μl of 2×sample buffer (EZ Apply, manufactured by ATTO) was added thereto, followed by stirring the resulting mixture using a vortex mixer, and heating the mixture in boiling water for 10 minutes to perform SDS treatment of the sample.

(106) (7) Purification of His-Tag Fusion Enzyme Protein from Bacterial Cell Fraction

(107) After thawing the bacterial cell fraction cryopreserved at −80° C., 50 ml of xTractor buffer (Clonthech) containing 5 μl of SEM nuclease (manufactured by Wako) was added thereto, and the resulting mixture was shaken for about 20 minutes at room temperature, to solubilize the bacterial cells. Similarly to the case of the supernatant sample, an EcoPack column (BIO RAD) was packed with 5 ml of TALON His-tag fusion protein purification resin (Clontech), and then purification of His-tag fusion enzyme protein was carried out by the the same method. Twenty microliters of each fraction was taken into a 1.5-ml Eppendorf tube, and 20 μl of 2×sample buffer (EZ Apply, manufactured by ATTO) was added thereto, followed by stirring the resulting mixture using a vortex mixer, and heating the mixture in boiling water for 10 minutes to perform SDS treatment of the sample.

(108) (8) Western Analysis of Purified Fractions

(109) As a standard substance for protein quantification, a purified esterase (EstA) preparation was used similarly to the case of quantification for the crude extract fraction. By 2-fold serial dilution of the preparation with a sample buffer, a dilution series to be used as standards was prepared. For electrophoresis (SDS-PAGE) of protein, an electrophoresis tank (Criterion cell, BIO RAD) and Criterion TGX-gel (BIO RAD) were used. In the electrophoresis tank, an electrophoresis buffer (Tris/Glycine/SDS Buffer, BIO RAD) was placed, and 4 μl of the SDS-treated fraction sample was applied to each well, followed by performing electrophoresis at a constant voltage of 200 V for 40 minutes.

(110) The gel after the electrophoresis was subjected to blotting by Trans-Blot Turbo (BIO RAD) using a Trans-Blot Transfer Pack (BIO RAD).

(111) The operation after the blotting was carried out by the same method as in the Western analysis of the crude extract sample described above.

(112) (9) Ultrafiltration and Concentration of Purified His-Tag Fusion Enzyme Protein

(113) The fractions for which the presence of the enzyme protein of interest could be confirmed by the Western analysis described above were combined together, and transferred to a centrifugal filtration unit (Amicon Ultra-10 K, Merck), followed by carrying out ultrafiltration at 5,000×g for 30 minutes at 4° C. Subsequently, the filtrate was discarded, and 10 ml of 1×PBS buffer (pH 7.4) was added, followed by carrying out ultrafiltration by the same method as described above. The same operation was further repeated three times, and the sample was concentrated to about 1.0 ml to remove imidazole in the sample. The enzyme protein concentration in the concentrate was determined by the Bradford method using BSA as a standard substance.

(114) (10) Measurement of Esterase Activity
(115) The activity of esterase was measured by the following method using a Lipase Kit S manufactured by DS Pharma Biomedical Co., Ltd.
(116) First, 2.4 ml of the included buffer was added to the coloring-agent container. After allowing complete dissolution, 22 ml of distilled water was added thereto to obtain a coloring stock solution. The coloring stock solution prepared was aliquoted in 1.1-ml volumes into 1.5-ml Eppendorf tubes, and cryopreserved at −20° C.
(117) Subsequently, the included reaction stop stock solution was incubated at 30° C. for 5 to 10 minutes to allow thawing, and the whole amount of the solution was diluted with distilled water to a final volume of 500 ml. The resulting dilution was stored in a refrigerator at 4° C. Immediately before the activity measurement, 1.1 ml of the coloring stock solution was thawed, and 1.1 ml of the included buffer and 8.8 ml of distilled water were added thereto, to prepare a coloring solution.
(118) Subsequently, for 100 μl of the coloring solution, 5 μl of the esterase sample for measurement was dispensed into a 96-well microplate, and the resulting mixture was mixed using a vortex mixer, followed by adding 10 μl of the included substrate solution (BALB) to each well, and mixing the resulting mixture. The plate was then covered with aluminum foil, and enzyme reaction was carried out at 30° C. for 30 minutes under dark conditions. Thereafter, 150 μl of the reaction stop solution, preliminarily incubated at 30° C., was added thereto to stop the reaction. As a blank, a sample series to which the substrate was added after the reaction was provided. The absorbances (412 nm) of both samples were measured using a microplate reader, and the value after subtraction of the blank value was multiplied by 1000 to obtain the BALB unit value, which was used as an index of enzyme activity.
(119) (11) Evaluation with Yeast *S. cerevisiae*
(120) By the method described in FIG. 2 of WO2017/115853, each transformed yeast, in which a plasmid encoding a fusion protein composed of human growth hormone protein (hGH) and a peptide tag added to the N-terminus thereof was introduced, was cultured, and 200 μl of the resulting culture was taken into a 1.5-ml Eppendorf tube. Thereafter, the tube was centrifuged at 2,000×g at 4° C. for 5 minutes, and then the supernatant was discarded, followed by freezing the precipitate (yeast cells) with liquid nitrogen and storing the precipitate at −80° C. Subsequently, the precipitate was thawed on ice, and the cell cluster was broken by vortexing. Thereafter, 0.5 ml of Yeast PreLysis buffer (manufactured by Atto Corporation) containing complete ULTRA cocktail (manufactured by Merck) and 0.1 μg/ml SEM nuclease (manufactured by Fujifilm Wako Pure Chemical Corporation) was added to the cells, and the cells were left to stand at room temperature for 10 minutes. Subsequently, the cell lysate was centrifuged at 10,000×g at 4° C. for 10 minutes to obtain the centrifugation supernatant and precipitate as a soluble fraction and insoluble fraction, respectively. After carrying out SDS treatment by a predetermined method, electrophoresis was carried out by SDS-PAGE, and then Western analysis was carried out to measure the amount of hGH in each fraction.
(121) <Results>
(122) (1) Improvement of Protein Expression Level in Soluble Fraction in *Brevibacillus* by Addition of Various Peptide Tags to N-Terminus
(123) The prepared recombinant *Brevibacillus* was cultured under predetermined conditions, and esterase was extracted therefrom under predetermined conditions, followed by measuring the expression level of the enzyme in the soluble fraction by Western analysis. As a result, as shown in FIG. **3**, when PG12 (SEQ ID NO:1) was added to the N-terminus in Comparative Example 2, no improvement of esterase expression was found in the soluble fraction relative to the case without addition of a tag. When PX12-20 was added to the N-terminus in Comparative Example 3, the expression level was 3.6 times higher than that in the case without addition of a tag. In contrast, when PX12-32, PX12-33, PX12-34, PX12-35, PX12-36, PX12-90, PX12-89, or PX12-83 was added to the N-terminus of esterase, expression levels not less than 6 times higher than that in the

case without addition of a tag were obtained, indicating that they are superior to the Comparative Examples. A particularly higher productivity was obtained when PX12-32 was added to the N-terminus.

(124) (2) Improvement of Protein Expression Level in Soluble Fraction in *Brevibacillus* by Addition of Various Peptide Tags to C-Terminus

(125) The prepared recombinant *Brevibacillus* was cultured under predetermined conditions, and esterase was extracted therefrom under predetermined conditions, followed by measuring the expression level of the enzyme in the soluble fraction by Western analysis. As a result, as shown in FIG. **4**, when PG12 (SEQ ID NO:1) was added to the C-terminus in Comparative Example 5, no improvement of esterase expression was found in the soluble fraction relative to the case without addition of a tag. When PX12-20 was added to the C-terminus in Comparative Example 6, the expression level was about 4 times higher than that in the case without addition of a tag. In contrast, when PX12-32 or PX12-34 was added to the C-terminus of esterase, expression levels not less than 8 times higher than that in the case without addition of a tag were obtained, clearly indicating that they result in higher production of esterase in the soluble fraction even compared to Comparative Example 6 (addition of PX12-20).

(126) (3) Influence of Addition of Various Peptide Tags on Esterase Activity

(127) The prepared recombinant *Brevibacillus* was cultured under predetermined conditions, and esterase secreted into the medium and esterase in the soluble fraction in the bacterial cells were purified under predetermined conditions, followed by investigation of the influence of the addition of each tag on the enzyme activity. As a result, as shown in FIG. **5**, N-terminal-tagged esterases secreted into the medium (PX12-32-S and PX12-34-S) were found to have the same activity as that of the non-tagged (Tag(−)) esterase. Further, N-terminal-tagged esterases obtained by homogenization of the bacterial cells (PX12-32-P and PX12-34-P) were also found to have the same activity as that of the non-tagged (Tag(−)) esterase.

(128) (4) Improvement of Protein Expression Level in Soluble Fraction in Yeast by Addition of Various Peptide Tags to N-Terminus

(129) The prepared recombinant yeast was cultured, and the expression level of hGH in each fraction was measured by Western analysis. As a result, as shown in FIG. **6**, when PX12-20-N was added to the N-terminus, the amount of hGH was smaller in the soluble fraction than in the insoluble fraction. However, when PX12-32 or PX12-34 was added to the N-terminus of hGH, the expression level of hGH in the soluble fraction remarkably increased. Thus, it was found that effective improvement of the protein expression level in the soluble fraction can be achieved also in eukaryotic cells.

(130) By the linking of a peptide tag containing a particular sequence, effective suppression of degradation of a fusion protein can be expected in eukaryotes.

Claims

1. A soluble fraction prepared by an expression system comprising a polynucleotide introduced in the expression system, wherein the polynucleotide encodes a fusion protein and comprises: a protein of interest; and a peptide tag linked to the protein of interest and comprising the following amino acid sequence:
X(PY).sub.qPZ, wherein P represents proline; X represents an amino acid sequence composed of 0 to 5 amino acids independently selected from the group consisting of arginine (R), glycine (G), serine(S), lysine (K), threonine (T), leucine (L), asparagine (N), glutamine (Q), and methionine (M); Y represents an amino acid sequence composed of 1 to 4 amino acids independently selected from the group consisting of G, T, Q, and M; q represents an integer of 1 to 10; and Z represents an amino acid sequence composed of 0 to 10 amino acids independently selected from the group consisting of R, G, S, K, T, L, N, Q, and M; with the proviso that the amino acid sequence of the

peptide tag comprises at least three Qs, Ms, Ls, Ns, and/or Ts in total; wherein the soluble fraction comprises the fusion protein produced and accumulated from the polynucleotide, and wherein PY is at least one selected from the group consisting of PGM, PGT, PQQ, PGMG, PGTG, and PQQQ.

2. The soluble fraction according to claim 1, wherein the peptide tag has a length of from 10 to 30 amino acids.

3. The soluble fraction according to claim 1, wherein the protein of interest is an enzyme.

4. The soluble fraction according to any one of claim 1, wherein the fusion protein comprises a secretion signal.

5. The soluble fraction according to claim 1, wherein the peptide tag is linked to a C-terminal side of the protein of interest.

6. A method of producing a fusion protein, the method comprising collecting the soluble fraction according to claim 1, and extracting the fusion protein.

7. The soluble fraction according to claim 1, wherein the peptide tag has the amino acid sequence of SEQ ID NO: 12, 15, 19, or 23.

8. A soluble fraction prepared by an expression system comprising a polynucleotide introduced in the expression system, wherein the polynucleotide encodes a fusion protein and comprises: a protein of interest; and a peptide tag linked to the protein of interest and comprising the amino acid sequence of SEQ ID NO: 12, 15, 19, 21, or 23; wherein the soluble fraction comprises the fusion protein produced and accumulated from the polynucleotide.